

I. Introduction & Comparison Overview

This report presents a comparative analysis of two system maps that model the relationship between economic autonomy, religion, social cohesion, and crime.

The two maps—Map A (Statistical Regression-Based Model) and Map B (Theory-Based Model with Hidden Features)—represent successive stages of analytical depth in understanding how social systems generate moral order.

Map A focuses on empirically testable relationships, using regression analysis to quantify the effects of economic and social variables on crime.

Map B, by contrast, extends this framework to include hidden systemic factors—such as cultural norms, policy support, and urban density—that shape social cohesion indirectly.

Together, they demonstrate a transition from a data-driven structural model (Week 4 regression) to a systemic, theory-grounded model (Week 5 systems thinking).

This evolution reflects the team's progression from isolated statistical associations to a complex adaptive understanding of social morality.

II. System Map Comparison

Map A – Regression-Based Statistical Model

Map A captures direct and measurable relationships between variables derived from the regression framework:

Economic Autonomy (X) → Social Cohesion (M) → Crime Rate (Y),
with **Religion Ratio (Z)** moderating the strength of the link between economic autonomy and social cohesion.

The model emphasizes quantitative clarity.

Each path corresponds to a regression coefficient (β) and R^2 value from empirical testing:

- Fiscal Independence → Crime ($\beta = -0.09$, weak direct effect)
- Religion Ratio → Crime ($\beta = -0.5$, significant negative effect)
- Volunteer Participation → Crime ($\beta = -1.3$, strong negative effect)
- Model Fit: **$R^2 = 0.73$**

Feedback interpretation: In this system, greater economic autonomy strengthens social infrastructure, which—when coupled with religious and civic engagement—reduces crime.

The primary feedback loop (B1) is **stabilizing**, as higher cohesion decreases crime, which further reinforces trust and participation.

Map B – Theory-Based Model (Including Hidden Features)

Map B builds upon Map A by integrating additional, **latent systemic factors** not directly captured in regression analysis.

It models a more realistic, multi-layered environment where social outcomes are influenced by economic, cultural, and policy-driven dynamics.

The structure expands as follows:

- **Economic Autonomy** influences **Social Cohesion**, which reduces **Crime Rate**.
- **Religion Ratio** moderates this pathway, strengthening the economic–social connection.
- Hidden variables—**Cultural Norms**, **Policy Support**, **Education Quality**, **Urban Density**, and **Media Influence**—act as **structural reinforcers** that stabilize or weaken the system.
- Feedback loops:
 - **R1 (Reinforcing Loop)**: Strong cohesion increases trust and community participation, further lowering crime.
 - **B1 (Balancing Loop)**: Excessive urban density or economic inequality reduces cohesion, offsetting R1.
 - **B2 (Policy Loop)**: Active policy support restores balance by rebuilding cohesion in stressed communities.

This map reveals the underlying **nonlinear interactions** and **feedback mechanisms** that statistical regression cannot fully capture—emphasizing complexity, adaptability, and systemic resilience.

III. Key Differences Table

Feature	Map A (Regression-Based)	Map B (Theory-Based)
Focus	Quantitative, empirical	Theoretical, systemic
Variables	X: Fiscal Independence M: Volunteer Rate Z: Religion Ratio Y: Crime Rate	Adds hidden variables: Cultural Norms, Policy Support, Education, Urban Density, Media Influence
Methodology	Multiple Linear Regression (OLS)	Systems Thinking / Causal Loop Modeling
Data Basis	Seoul district-level dataset (n=25)	Theoretical synthesis from literature and context
Feedback Loops	Single balancing loop (Cohesion ↓ Crime ↓)	Multiple reinforcing and balancing loops (R1, B1, B2)
Interpretive Strength	High empirical clarity, limited scope	High systemic realism, less measurable
Outcome	Crime reduction explained ($R^2 = 0.73$)	Broader explanation of moral system resilience

IV. Justification

1. Why certain 'non-significant' variables might still matter

Although fiscal independence showed a weak statistical link to crime ($\beta \approx -0.09$, $R^2 = 0.13$), it plays a structural enabling role within the system.

Economic autonomy provides the foundational stability—employment, welfare, and equity—that allows social cohesion to flourish.

In Map B's systemic interpretation, fiscal autonomy is not a direct cause but a latent stabilizer, maintaining the environment where trust and volunteerism can sustain over time.

In complex adaptive systems, such "non-significant" variables often act as slow variables—drivers of long-term equilibrium that remain invisible in short-term regressions. Their influence unfolds through feedback mechanisms rather than direct linear effects.

2. Policy implications if we ignore these hidden features

Relying solely on statistically significant predictors like religion or volunteerism risks short-sighted policy design.

Ignoring economic and cultural structures can lead to interventions that appear effective but quickly degrade once funding or public engagement declines.

Hidden variables—policy support, education, urban density, and cultural norms—ensure that social cohesion is maintained under pressure.

Neglecting these creates vulnerability to systemic breakdown, where moral norms weaken despite temporary community programs.

Therefore, policies should operate on two levels:

1. Strengthen observable drivers (volunteering, participation);
2. Reinforce hidden foundations (economic autonomy, inclusive governance, cultural capital).

Ultimately, not all significance is statistical.

Some variables act as the invisible scaffolding that keeps moral and social systems standing.

V. Executive Insight

Statistical insignificance does not mean social irrelevance.

Hidden variables form the invisible architecture of moral order, binding economic, cultural, and social systems into one cohesive whole.